Exact replication and screening of tropical finiteness certificates for central configurations

Machine record v0.02 (draft): the CAOS Research central-configurations program, experiments 001–005

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Abstract

We report an exact, fully replayable machine record around Smale's 6th problem (finiteness of planar central configurations). Contributions of this record: (i) an exact calibration of the Albouy–Chenciner mutual-distance equations in open tooling, including a machine rediscovery of their dimension-blindness (the regular tetrahedron coexists with the square in the equal-mass rhombus stratum of the 4-body distance system until the Cayley–Menger equation is adjoined, after which the square, with side a satisfying $32a^6 - 32a^3 + 7 = 0$, is the unique positive point of the stratum); (ii) a machine measurement of the *enrichment law*: adjoining the asymmetric (Roberts) equations and the energy–inertia relation renders the $n = 3$ distance ideal zero-dimensional directly, eliminating the coordinate-line components of the bare symmetric system; (iii) an independent, exact reproduction of the Jensen–Leykin tropical prevariety certificate of generic-mass finiteness for $n = 5$ (arXiv:2301.02305v2): both published f-vectors match digit for digit and the pointedness of the recession cone is verified by an independent exact parser; and (iv) a screening study over mass-valuation families and equation-system variants at $n = 4, 5$ which finds two new working valuation families beyond the two published, replicates generic-mass finiteness for $n = 4$ purely polyhedrally in seconds, and shows that removing the ten dependent symmetric Albouy–Chenciner equations destroys the certificate at every tested valuation: the "redundant" equations carry essential tropical information. All computations are exact (rational or integer arithmetic, or certified polyhedral computation); every stated number is reproducible from committed code and hashes. This is a replication-and-instruments record: the theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin, and Jensen–Leykin; the measurements in (i), (ii) and (iv) are, to our knowledge, recorded here at this level of explicitness for the first time.

1 Introduction

A *central configuration* (CC) of the Newtonian n-body problem with masses $m_1, \ldots, m_n > 0$ is a configuration $x = (x_1, \ldots, x_n)$, $x_i \in \mathbb{R}^d$, satisfying

$$\lambda\,(x_i - c) = \sum_{j \neq i} \frac{m_j (x_j - x_i)}{r_{ij}^3}, \qquad i = 1, \ldots, n, \tag{1}$$

*With machine-executed computations; all artifacts, exact values, and run records are versioned at `https://github.com/fsantibanezleal/CAOS_RESEARCH` under `problems/dynamical-systems/central-configurations/`.

for some λ, where c is the center of mass and $r_{ij} = |x_i - x_j|$. Smale's 6th problem for the 21st century [1] asks, following Chazy [2] and Wintner [3], whether for every n and every choice of positive masses the number of *planar* central configurations, up to symmetry, is finite. The modern ladder is: $n = 4$ finite for all positive masses (Hampton–Moeckel [4]: computer-assisted BKK certificate, count in $[32, 8472]$); $n = 5$ finite except possibly on an explicit codimension-two subvariety of mass space (Albouy–Kaloshin [5]); the spatial 5-body Dziobek case handled generically by Moeckel [6] and explicitly by Hampton–Jensen [7]; $n = 6$ open, reduced by Chang–Chen [8] to a finite list of residual diagram cases; and, for *generic* masses, the recent purely polyhedral certificate of Jensen–Leykin [9], complete through $n = 5$. Rigorous equal-mass censuses for $n \leq 7$ are due to Moczurad–Zgliczyński [10]; with one negative mass a continuum exists (Roberts [11]), so positivity is essential.

This paper is a *machine record*: it documents, with exact artifacts, a replication-first program whose goal is to place every certificate of the ladder inside one open, replayable toolchain, and then to run the experiments the frontier papers themselves call for. We follow a strict discipline: hypotheses are declared and committed before every run; verdicts record refutations as prominently as confirmations; every quantitative claim below is reproducible from the versioned repository, and the git and Zenodo timestamps constitute the priority record of the work.

2 Exact calibration of the Albouy–Chenciner system (EXP-001)

Working in the mutual distances r_{ij} with the scale normalization $S_{ij} = r_{ij}^{-3} - 1$ ($S_{ii} = 0$), the symmetric Albouy–Chenciner (AC) equations [12, 4] are, after clearing denominators,

$$f_{ij} = \sum_{k=1}^{n} m_k \left[S_{ik} \left(r_{jk}^2 - r_{ik}^2 - r_{ij}^2 \right) + S_{jk} \left(r_{ik}^2 - r_{jk}^2 - r_{ij}^2 \right) \right] = 0. \tag{2}$$

Our exact implementation (sympy over $\mathbb{Q}$) reproduces, for four exact mass vectors, the classical $n = 3$ landscape: the Lagrange point $r_{12} = r_{13} = r_{23} = 1$ identically in symbolic masses, and exactly one positive collinear solution per ordering (Euler–Moulton; for equal masses the 2-middle chart value is $r_{12} = r_{23} = \sqrt[3]{90}/6$), with the degree-54 symbolic Euler eliminant persisted. For $n = 4$ each cleared AC polynomial has exactly 14 monomials, matching the count stated in [4].

The instructive result of the calibration is a machine *refutation* of a declared prediction. In the equal-mass rhombus stratum $r_{12} = r_{23} = r_{34} = r_{14} = a$, $r_{13} = r_{24} = b$ of $n = 4$, the bare AC system has *two* positive solutions: the planar square, with $b^2 = 2a^2$ and $a^3 = (4 + \sqrt{2})/8$ (minimal polynomial $32a^6 - 32a^3 + 7$), and the regular tetrahedron $a = b = 1$, whose bordered Cayley–Menger determinant equals $4 \neq 0$. The distance equations are dimension-blind, exactly as stated in [4]; planarity is restored only by adjoining the Cayley–Menger equation, after which the square is the unique positive point of the stratum. We record the scale-free invariant $J = UI^{1/2}/M^{5/2}$ at both anchors: $J_\square = \frac{1}{16} + \frac{\sqrt{2}}{8}$, $J_{\text{tet}} = \frac{3\sqrt{6}}{32}$.

Two toolchain findings of independent interest were caught and are now regression-gated in the repository: `sympy`'s `solve_poly_system` silently returns incomplete solution lists on these systems (it missed the square, whose eliminant factor $4b^6 - 4b^3 - 7$ is even radical-solvable), and `is_positive` returns `None` on nested algebraic numbers, so naive filters silently drop genuine solutions. Verdict-grade counting in our pipeline therefore uses univariate eliminants that are provably elements of the ideal (lex Gröbner elimination, or Stickelberger multiplication-matrix characteristic polynomials), complete real root isolation, and exact residual acceptance.

3 The enrichment law (EXP-002)

The bare symmetric system is never zero-dimensional: the line $r_{13} = r_{23} = 0$ lies in its variety for all masses (machine-verified symbolically), and saturation is expensive. Adjoining the $n(n-1)$ asymmetric equations $g_{ij} = \sum_k m_k S_{ik}(r_{jk}^2 - r_{ik}^2 - r_{ij}^2)$ (observed by G. Roberts; see [7, 9]) and the energy–inertia relation $U = MI$ changes the picture completely: for all four sample mass vectors the enriched ideal is zero-dimensional *directly* (Gröbner basis of size 24 in under a second each), and the full positive censuses, where decided, are exactly classical: four points (one equilateral, three collinear), with no spurious solutions. The planar rhombus census on the enriched system with Cayley–Menger adjoined yields the square alone. This is the affine-level instance of a pattern that recurs at every level of the subject: Hampton–Moeckel could not run their method on the six AC equations alone and required Dziobek's equations [4]; Hampton–Jensen kept the asymmetric equations to refine their tropical cones [7]; and Jensen–Leykin explicitly call for "different versions of the Albouy–Chenciner equations" at $n = 6$ [9]. Section 5 quantifies this pattern at the prevariety level.

4 Exact reproduction of the Jensen–Leykin $n = 5$ certificate (EXP-003)

Jensen–Leykin [9] prove generic-mass finiteness of planar CCs for $n \leq 5$ by a purely polyhedral computation: specialize the masses into the Puiseux field with chosen valuations, compute the tropical prevariety of the system (20 asymmetric AC + 10 symmetric AC + 5 planar Cayley–Menger polynomials for $n = 5$) with `gfan` 0.7, and verify that every connected component has a pointed recession cone; Bieri–Groves and the density of tropicalization fibers then yield finiteness for almost all masses.

We rebuilt `gfan` 0.7 from the author tarball (one two-line `#include <cstdint>` patch for gcc 13; tarball and binary SHA-256 recorded) and ran the paper's own command pipeline. Results: with valuations $(1, 3, 9, 27, 81)$ the prevariety f-vector is $(1506, 4744, 8586, 8787, 4652, 993)$, and among its 85 unbounded ray directions none has a positive coordinate (verified by our independent exact parser): the globally pointed certificate, exactly as published. With valuations $(1, 4, 9, 16, 25)$ the f-vector is $(3586, 12012, 18531, 15625, 7072, 1357)$, again matching [9] digit for digit. Each run took under six minutes of wall time on 30 threads ($\approx$ 145 cpu-minutes). To our knowledge this is the first published independent replication of the [9] computation; at the measured throughput, a single $n = 6$ attempt at the scale they report ($\approx$ 100 cpu-days) costs about three wall-days on commodity hardware.

5 Valuation and equation-variant screening (EXP-004)

The certificate of [9] has two designed inputs: the mass valuations and the equation system. The paper publishes two working valuation choices for $n = 5$ and, for the inconclusive $n = 6$ attempt, asks explicitly for "more experimentation with different valuations or different versions of the Albouy–Chenciner equations". We screened both axes on a 16-cell grid: two systems (S1, the full set of asymmetric + symmetric AC equations + planar Cayley–Menger determinants; S2, the same with the $\binom{n}{2}$ dependent symmetric equations removed) crossed with six valuation families at $n = 5$, plus four cells at $n = 4$. Pointedness is decided per connected component (*comet*) by an exact certificate: a rational vector c with $c \cdot r < 0$ for every recession generator r, verified in rational arithmetic (a floating-point solver only proposes c).

Valuations at $n = 5$ (system S1)	comets	pointed	status
$(1, 3, 9, 27, 81)$ powers of 3	281	281	certificate (globally pointed)
$(1, 4, 9, 16, 25)$ squares	257	257	certificate
$(1, 2, 4, 8, 16)$ powers of 2	266	266	certificate (new)
$(2, 3, 5, 7, 11)$ primes	250	250	certificate (new)
$(0, 1, 2, 3, 4)$ arithmetic	181	180	no certificate (1 comet undecided)
$(1, 1, 9, 27, 81)$ repeated	211	210	no certificate (1 comet undecided)

Table 1: Comet-level screening at $n = 5$. The squares row reproduces the 257 components reported in [9], a quantity independent of the f-vector, which validates the component extraction as well as the polyhedral computation.

Three findings.

(a) Two new working valuation families. Powers of two and primes both certify at $n = 5$; before this screening, two families were known to work. Since the certificate is a statement about one point of the valuation space, each working family is an independent verification of generic finiteness for $n = 5$, and the enlarged pool is what makes an $n = 6$ attempt a design choice rather than a lottery. This matters operationally: powers of three reach t^{243} at $n = 6$ and exceed `gfan`'s machine-integer fast path (our first $n = 6$ attempt aborted on a detected overflow after six minutes), whereas powers of two reach only t^{32}.

(b) The dependent symmetric equations are essential. Every S2 cell has a strictly larger f-vector in every entry than its S1 counterpart and yields no certificate at any valuation, including the one where S1 is globally pointed; the single undecided comet grows from 71–95 recession generators in the S1 controls to 241–2224 in S2. The equations that are algebraically redundant (each symmetric equation is the sum of two asymmetric ones) are tropically load-bearing. This is the prevariety-level form of the affine phenomenon of Section 3, and it agrees with the historical record: [4] could not run their method on the six AC equations alone, and [7] kept the redundant family deliberately.

(c) $n = 4$ falls in seconds, and the hard case is exactly the predicted one. With valuations $(1, 3, 9, 27)$ and $(1, 2, 4, 8)$ the $n = 4$ prevariety is globally pointed (10 comets each; 4 and 2 seconds), a purely polyhedral replication of generic-mass finiteness for the four-body problem; arithmetic valuations $(0, 1, 2, 3)$ also certify (9 comets). The equal-valuation case $(0, 0, 0, 0)$ leaves a single undecided comet with 49 generators, matching the observation of [9] that this specialization is equivalent in difficulty to the original computation of [4].

We are careful about the negative half: our instrument certifies pointedness (exact separating vector) and unpointedness (exact nonnegative zero combination); on the failing controls it returns neither, so we report "no certificate" rather than "not pointed". Closing that gap with exact linear programming is a queued item.

6 The $n = 6$ attempt (EXP-005, in progress)

The screening of Section 5 turns the $n = 6$ question into a design problem with two dials. Our first attempt (powers of three, machine-integer fast path) aborted after six minutes on a *detected* overflow, which is the benign failure mode: `gfan` raises rather than returning a wrong answer. Two variants now run in parallel on the same hardware: powers of two with the fast path (maximum exponent 32 instead of 243) and powers of three with arbitrary-precision integers, the slower path described in [9]. At the throughput measured in Section 4, an attempt at the scale they report costs on the order of days rather than weeks. The outcome, positive or negative, will be reported with its exact certificates in the next version of this record; a positive outcome would extend generic-mass finiteness to $n = 6$, and a negative one localizes the obstruction to specific comets, data that has not been published for any failed attempt.

7 Honesty ledger

This record is replication-first. The finiteness theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin, and Jensen–Leykin; the equal-mass censuses to Moczurad–Zgliczyński. Our measurements (the dimension-blindness anchors with their exact minimal polynomials and J-invariants; the enrichment law at affine and prevariety level; the screening tables) are machine facts about known objects, stated with their complete provenance. Open items are tracked in the repository backlog; in particular, censuses for two integer-separated mass vectors are inconclusive under our current sympy engine caps and are being re-run under an independent engine (`msolve`) with the sympy layer retained as verification. No claim in this paper depends on an unverified literature statement; the repository flags every such statement as UNVERIFIED until read from its primary source.

References

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